Evolutionary Computation for Tax-Minimizing Strategies in Special Economic Zones

DAG Structures for Testing

Structures, Metrics, and Economic Rationale

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Abstract

This document compiles a library of Directed Acyclic Graphs (DAGs) tailored for model testing. Each entry includes the figure, a compact set of structural metrics (node and edge counts, number of layers, longest path as a depth proxy, and density), and an accompanying economic interpretation. Our aim is twofold: (i) to offer reproducible synthetic topologies that isolate features we believe may affect auditability (e.g., split–merge diamonds, inter-layer cross-edges, hubs, and late-stage convergence); and (ii) to provide narrative rationales that link those features to practical risks.

Unless otherwise noted, the number of nodes is held fixed at $N = 15$ to enable like-for-like comparisons across structures. The catalog is intended as a reference during scenario design and as a didactic artifact for onboarding new analysts. The testbed currently covers: All Inputs, Assembly Tree, Butterfly, Dominant Input, Explosive Expansion, Hyper-Dense, Interlaced Pathway, Linear, Linear All-Inputs, Mangrove, Sequential Mergers with External Inputs, Successive Diamonds, Tri-Cluster, and Tri-Spine Convergent.

How to Read

Layout. All graphs are DAGs laid out from sources (top) to sinks (bottom). Layers approximate production stages.

Metrics. We report: *Number of Nodes*, *Number of Edges*, *Max Betweenness Centrality (MBC)*, *Longest Path*, *Density*, *Number of Production Decisions*, and *Search Space Size*. These are comparable across cases.

Text fields. Each page contains a two-paragraph note: *Economic structure* (what the topology looks like) and *Justification* (why that structure matters for prices, margins, and traceability).

Navigation. Use the table of contents to jump to any graph; the consolidated summary at the end links back to each page.

Node count. Unless otherwise noted, all graphs keep the number of nodes fixed at $N = 15$ to ensure comparability across topologies.

Search Space Complexity and Heuristic Necessity

While assigning binary zone designations over a strictly linear sequence yields a modest search space (e.g., 216=65,536) that is computationally trivial for exact solvers, the scope of this framework extends far beyond such simplified configurations. If the model were restricted exclusively to linear chains, heuristic methods would indeed be unnecessary.

However, the combinatorial optimization problem defines its decision space based on the number of activities or decisions (n) required to execute the production graph, rather than merely the raw node count. As demonstrated in Table 1, when intermediate goods are reused, or when topologies involve parallel processing and multiple cross-layer links (e.g., the Hyper-Dense or Butterfly graphs), the number of decisions n explodes. This non-linear scaling generates search spaces ranging from computationally manageable to strictly intractable.

The Genetic Algorithm was therefore implemented to ensure that the anticipatory framework remains robust, scalable, and topologically agnostic. This heuristic approach is a mathematical necessity for evaluating the highly interconnected production networks typical of real-world multinational structures, where exact enumeration is impossible.

Table 1: Summary of Structural Metrics and Search Space Complexity per Topology

Topology Name	**Nodes**	**Edges**	**MBC**	**Path**	**Density**	**Decisions** (n)	**Search Space Size** (2^n)
All Inputs	15	14	0	1	0.133	**16**	2^{16} **(32,768)**
Linear	15	14	0.269	14	0.133	**16**	2^{16} **(65,536)**
Linear All-Inputs	15	27	0.230	14	0.257	**16**	2^{16} **(65,536)**
Seq. Mergers / Ext. Inputs	15	14	0.131	7	0.133	**16**	2^{16} **(65,536)**
Tri-Cluster	15	18	0.071	4	0.171	**20**	2^{20} ($\approx 1.04 \times 10^6$)
Tri-Spine Convergent	15	15	0.049	5	0.143	**20**	2^{20} ($\approx 1.04 \times 10^6$)
Dominant Input	15	27	0.230	14	0.257	**29**	2^{29} ($\approx 5.36 \times 10^8$)
Explosive Expansion	15	20	0.036	4	0.190	**30**	2^{30} ($\approx 1.07 \times 10^9$)
All-in-one	15	19	0.065	5	0.180	**33**	2^{33} ($\approx 8.58 \times 10^9$)
Assembly Tree	15	25	0.030	4	0.238	**44**	2^{44} ($\approx 1.75 \times 10^{13}$)
Successive Diamonds	15	22	0.075	5	0.209	**50**	2^{50} ($\approx 1.12 \times 10^{15}$)
Mangrove	15	31	0.074	5	0.295	**72**	2^{72} ($\approx 4.72 \times 10^{21}$)
Interlaced Pathway	15	23	0.113	12	0.219	**138**	2^{138} ($\approx 3.48 \times 10^{41}$)
Butterfly	15	26	0.115	7	0.247	**256**	2^{256} ($\approx 1.15 \times 10^{77}$)
Hyper-Dense	15	92	0	5	0.876	**769**	2^{769} ($\approx 3.10 \times 10^{231}$)

1 All Inputs

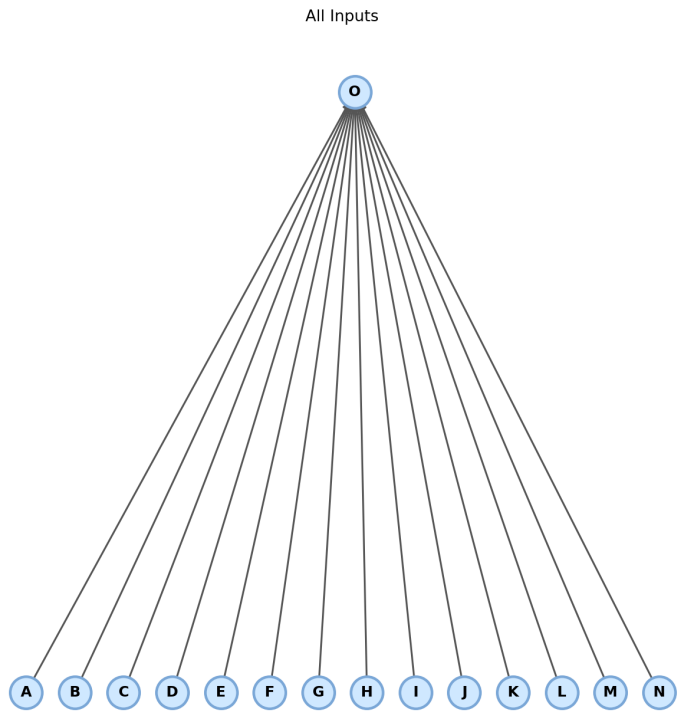

Figure 1: All Inputs

Structure. A single final assembler (O) receives inputs directly from numerous independent producers, forming a pure star topology where all production paths converge in one step without intermediate processing.

Justification. This configuration represents a centralized assembly or procurement structure—common in large holding companies or anchor firms sourcing from multiple subsidiaries. It serves as a baseline case to evaluate how profit recognition concentrates at the central node and how limited inter-firm trade reduces, but does not eliminate, opportunities for cost misallocation or transfer-price manipulation through direct supplier relationships.

Core metrics

Metric	Value
Number of Nodes	15
Number of Edges	14
Max Betweenness Centrality	0
Longest Path	1
Density	0.133

2 Assembly Tree

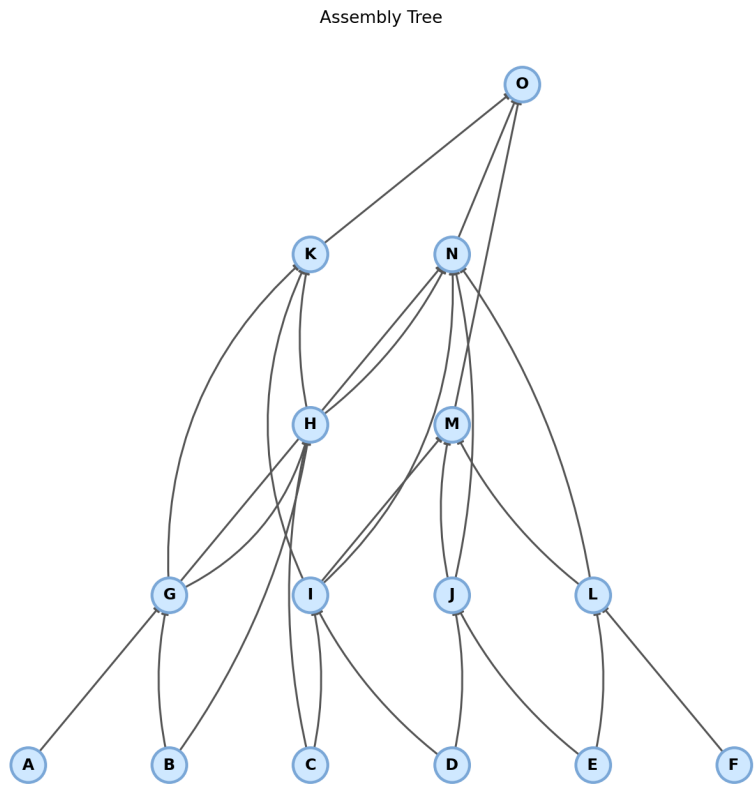

Figure 2: Assembly Tree

Economic structure. A branched, two-layer assembly that converges to a single final good O. Upstream sources (A. . . F) feed mid-tier modules (G,H,I,J,L), which then assemble into late modules (K,M,N) before reaching O. It's tree-like (arborescent) with limited cross-links that preserve acyclicity.

Justification. Multiple assembly tiers create several points to recognize taxable income on inter-company transfers. Nodes like G (highest betweenness) mediate many shortest routes to O even under complexx structures; small mark-up changes there influence a large share of downstream profit, making this instance ideal for testing strategies.

Core metrics

Metric	Value
Number of Nodes	15
Number of Edges	25
Max Betweenness Centrality	0.03
Longest Path	4
Density	0.238

3 Butterfly

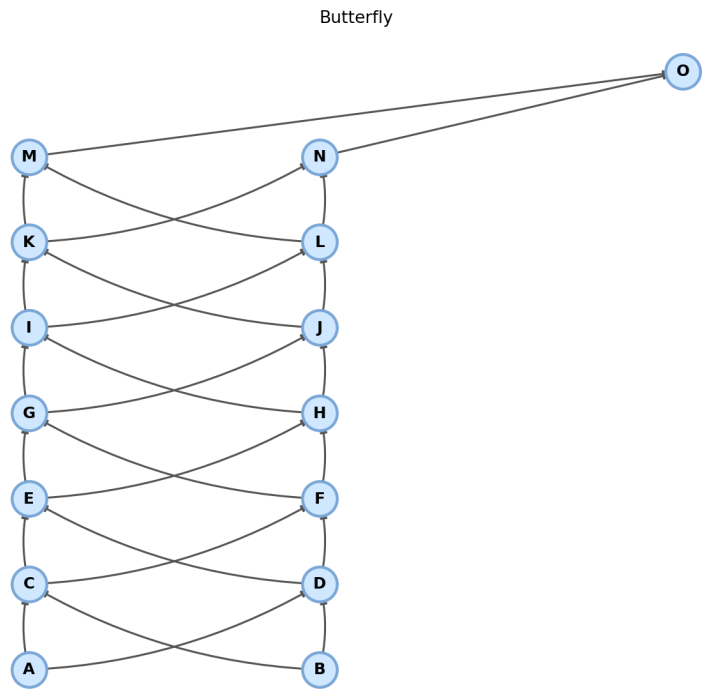

Figure 3: Butterfly

Economic structure. Every stage contains a full bipartite connection to the next, producing a dense, symmetric grid of inter-firm dependencies. Each node at a given tier can source inputs from multiple predecessors and supply multiple successors, mimicking a tightly integrated production ecosystem.

Justification. This configuration reproduces complex value chains where goods and capital circulate through overlapping ownership structures. The redundancy of routes allows firms to reassign production flows and internal prices across many equivalent paths, enabling high flexibility for cost shifting and margin reallocation—conditions under which profit-shifting detection becomes especially challenging.

Core metrics

Metric	Value
Number of Nodes	15
Number of Edges	26
Max Betweenness Centrality	0.115
Longest Path	7
Density	0.247

4 Dominant Input

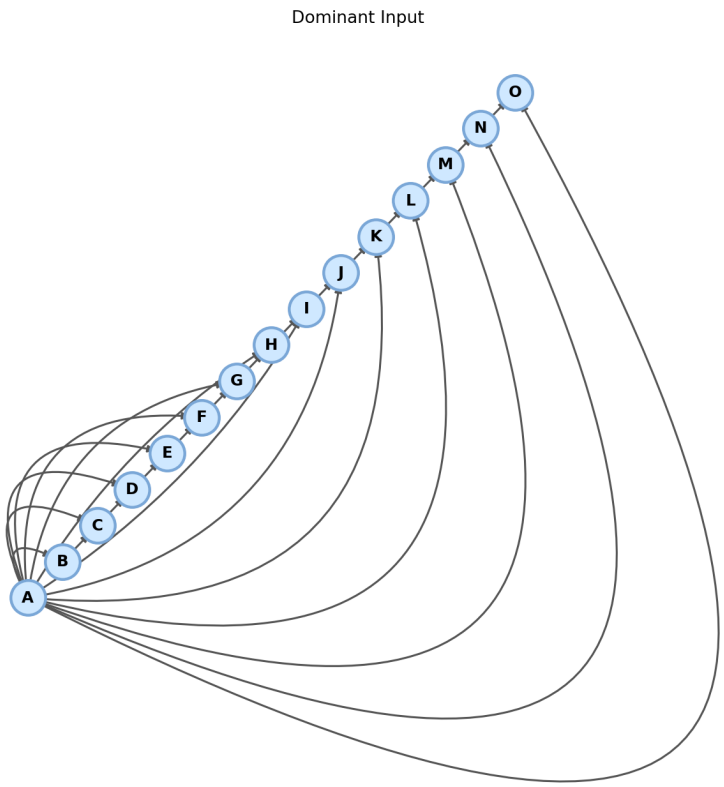

Figure 4: Dominant Input

Economic structure. A single dominant input (A) feeds every stage of production while a secondary linear chain (B→C→. . . →O) propagates sequential transformations. The structure mixes a hierarchical dependency on one key input with a serial progression of value addition.

Justification. This configuration represents industries centered around a pivotal raw material or proprietary asset that participates in all production stages—such as technology, licensing, or energy inputs. It allows the model to test how dependency on a dominant supplier shapes internal pricing and how control over a critical input can be used to shift profits along the production chain through strategic transfer pricing.

Core metrics

Metric	Value
Number of Nodes	15
Number of Edges	27
Max Betweenness Centrality	0.23
Longest Path	14
Density	0.257

5 Explosive Expansion

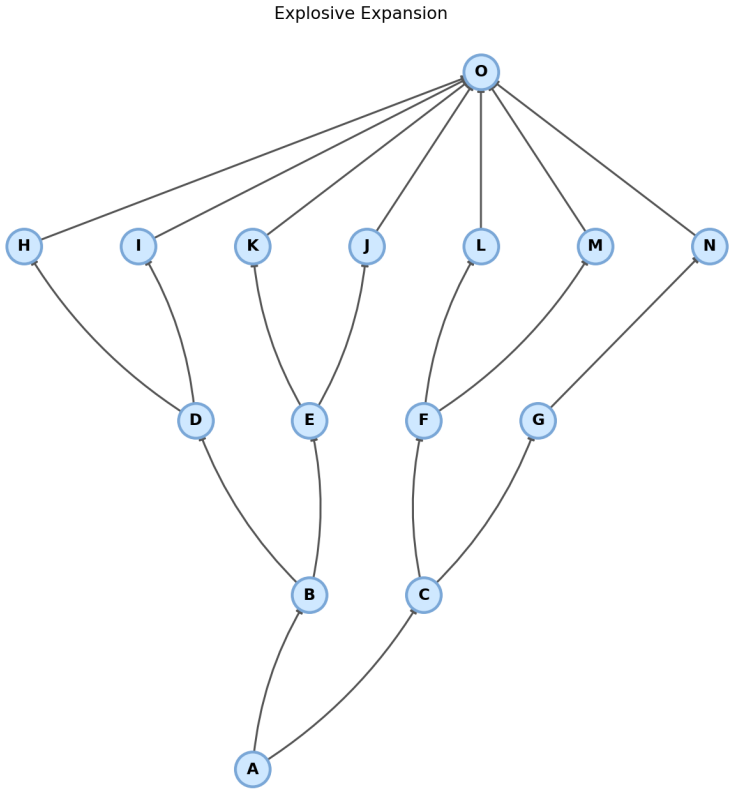

Figure 5: Explosive Expansion

Economic structure. A branching hierarchy where each production stage multiplies into several independent subpaths that ultimately converge in a single final output. The process expands rapidly from a few initial inputs into a wide array of intermediates before collapsing back into one assembler.

Justification. This structure represents industries with high product diversification and distributed intermediate production—such as component manufacturing or modular assembly chains. The rapid branching amplifies the number of entities and internal transactions, providing fertile ground for hidden mark-ups and complex transfer-pricing patterns as profits can be dispersed among many nodes before consolidation.

Core metrics

Metric	Value
Number of Nodes	15
Number of Edges	20
Max Betweenness Centrality	0.036
Longest Path	4
Density	0.19

6 Hyper-Dense

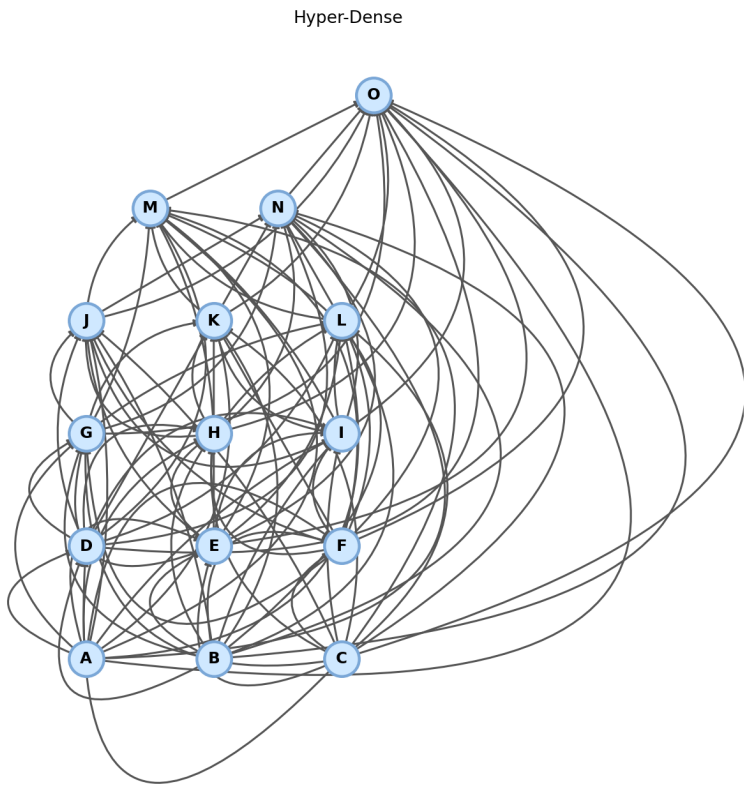

Figure 6: Hyper-Dense

Economic structure. An almost fully interconnected hierarchy where each layer of firms maintains trade links with nearly all nodes in subsequent layers. Multiple overlapping routes connect every stage to the final output, creating extreme redundancy and interdependence across the entire system.

Justification. This highly connected configuration emulates a conglomerate-like economy in which production, sourcing, and assembly occur through a dense web of internal exchanges. Such density amplifies the degrees of freedom for price manipulation, cost reallocation, and profit-shifting—making it an ideal stress test for the model's ability to detect evasive behaviors in environments where traditional audit trails lose clarity due to excessive interlinkages.

Core metrics

Metric	Value
Number of Nodes	15
Number of Edges	92
Max Betweenness Centrality	0
Longest Path	5
Density	0.876

7 Interlaced Pathway

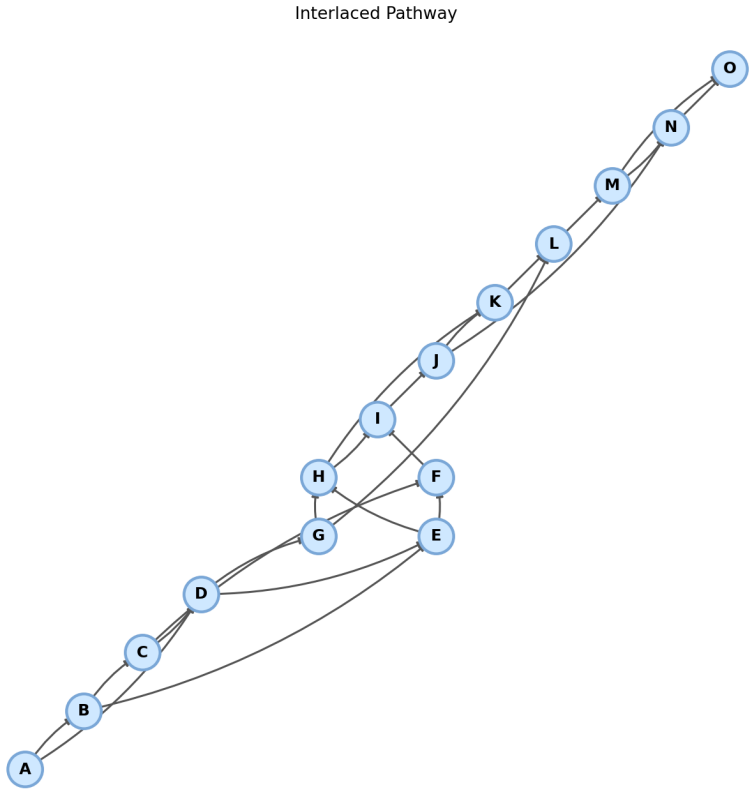

Figure 7: Interlaced Pathway

Economic structure. A deep, asymmetrical hierarchy where multiple partially overlapping chains form a dense web of forward and backward interconnections. Parallel paths repeatedly merge and diverge, producing a structure that balances hierarchy and lateral exchange.

Justification. This configuration resembles a multi-division production ecosystem with recursive sourcing between intermediate stages. Firms at similar levels can transact both vertically and horizontally, allowing for strategic reallocation of profits through reciprocal pricing.

Core metrics

Metric	Value
Number of Nodes	15
Number of Edges	23
Max Betweenness Centrality	0.113
Longest Path	12
Density	0.219

8 Linear

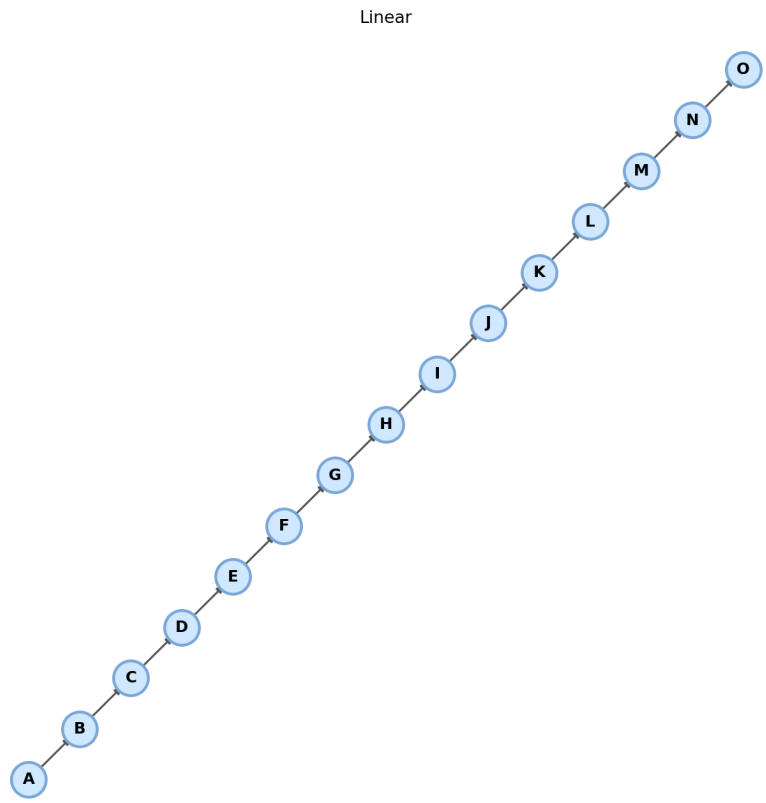

Figure 8: Linear

Economic structure. A single path A→B→→O with one final good (O).

Justification. Many profit-recognition points: 14 intermediate transfers create multiple places to recognize taxable income via internal prices of semi-finished goods.

Jurisdictional allocation: mapping some stages to FTZ and others to NCT lets the chain reallocate income along the spine via mark-up choices, without needing parallel routes.

Core metrics

Metric	Value
Number of Nodes	15
Number of Edges	14
Max Betweenness Centrality	0.269
Longest Path	14
Density	0.133

9 Linear All-Inputs

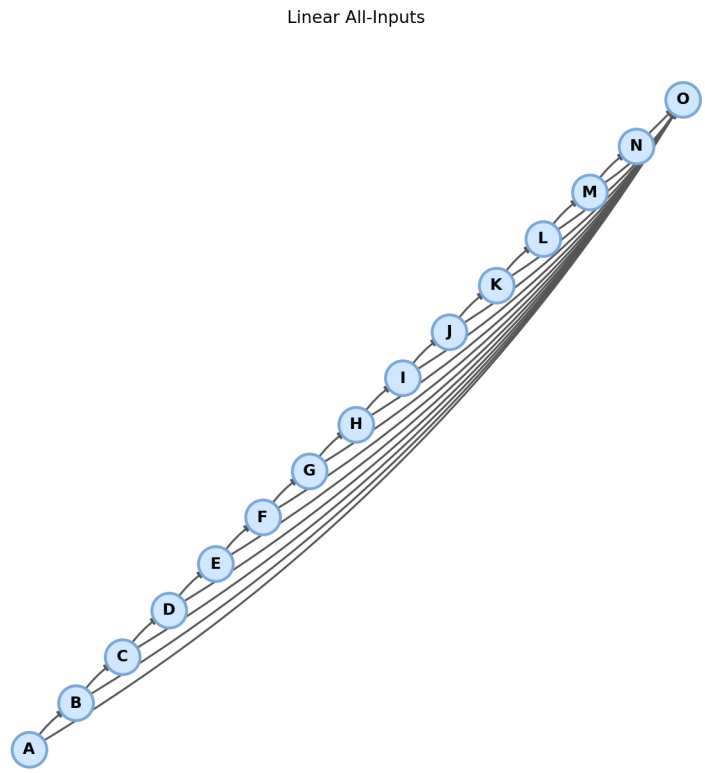

Figure 9: Linear All-Inputs

Economic structure. A deep path A→B→→O plus direct arcs from any intermediate good to the final good O. This encodes a "close-to-final assembly" option at almost every stage: one can either keep transforming along the chain or feed the partially processed good directly into O

Justification. Multiple links to O let the producer decide where along the chain to "close" value into the final good. Setting transfer prices at those exit edges reallocates taxable income across entities/stages without changing the physical DAG.

If some upstream stages are in FTZ and others in NCT, choosing the exit from an FTZ-located stage with a higher mark-up shifts income toward FTZ while keeping a production route to O.

Core metrics

Metric	Value
Number of Nodes	15
Number of Edges	27
Max Betweenness Centrality	0.23
Longest Path	14
Density	0.257

10 Mangrove

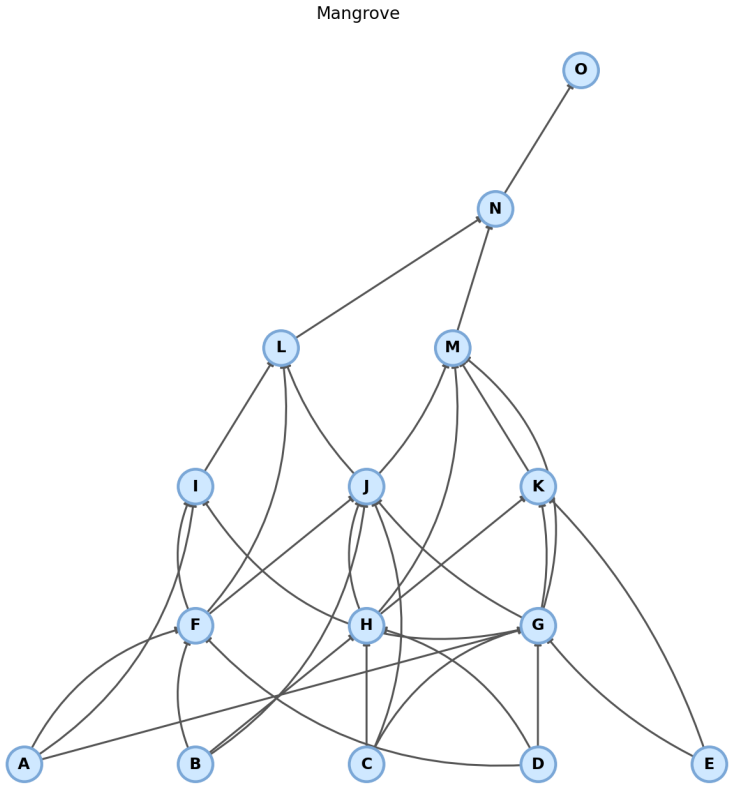

Figure 10: Mangrove

Economic structure. Interwoven roots of partially overlapping production chains converge and diverge repeatedly before reaching the final output. Multiple upstream nodes feed shared intermediates, creating loops of partial dependency that resemble the tangled structure of a mangrove.

Justification. This graph captures the dynamics of inter-firm production systems where inputs and semi-finished goods circulate across overlapping branches. Such entanglement generates multiple plausible transfer routes and ownership boundaries, allowing firms to redistribute profits across interconnected subsidiaries while maintaining operational legitimacy—making it a powerful test case for tracing concealed profit flows within dense, semi-circular supply webs.

Core metrics

Metric	Value
Number of Nodes	15
Number of Edges	31
Max Betweenness Centrality	0.074
Longest Path	5
Density	0.295

11 Sequential Mergers with External Inputs

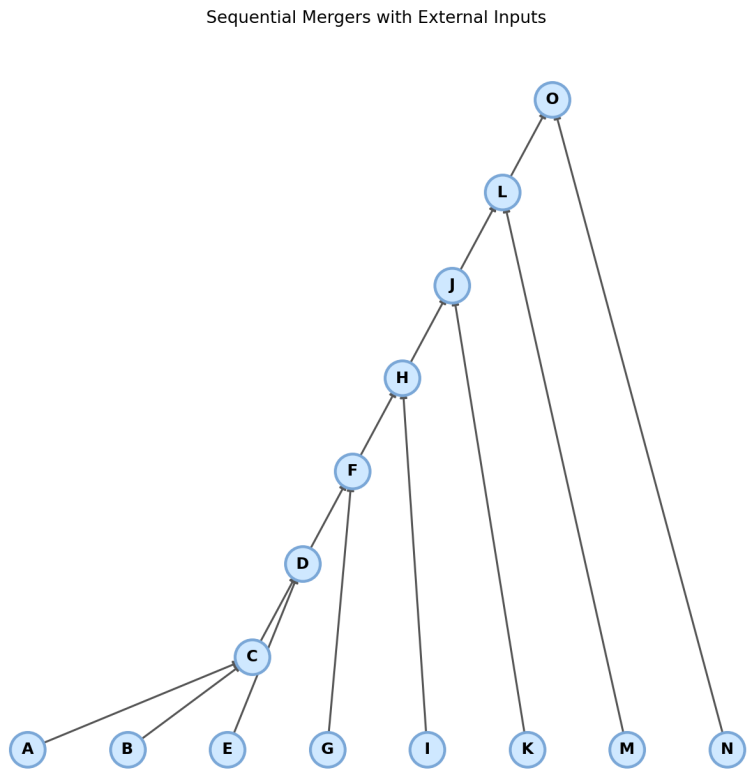

Figure 11: Sequential Mergers with External Inputs

Economic structure. Each production stage merges one internally generated good with a new external input purchased from the market, forming a chain of successive integrations that progressively build toward the final product.

Justification. Because every stage requires an additional good sourced from outside the current production line, firms must engage in repeated market transactions or related-party purchases. This creates a layered structure where production costs and transfer prices intertwine, offering realistic conditions for testing how external sourcing amplifies opportunities for profit-shifting and cost reallocation across entities.

Core metrics

Metric	Value
Number of Nodes	15
Number of Edges	14
Max Betweenness Centrality	0.131
Longest Path	7
Density	0.133

12 Successive Diamonds

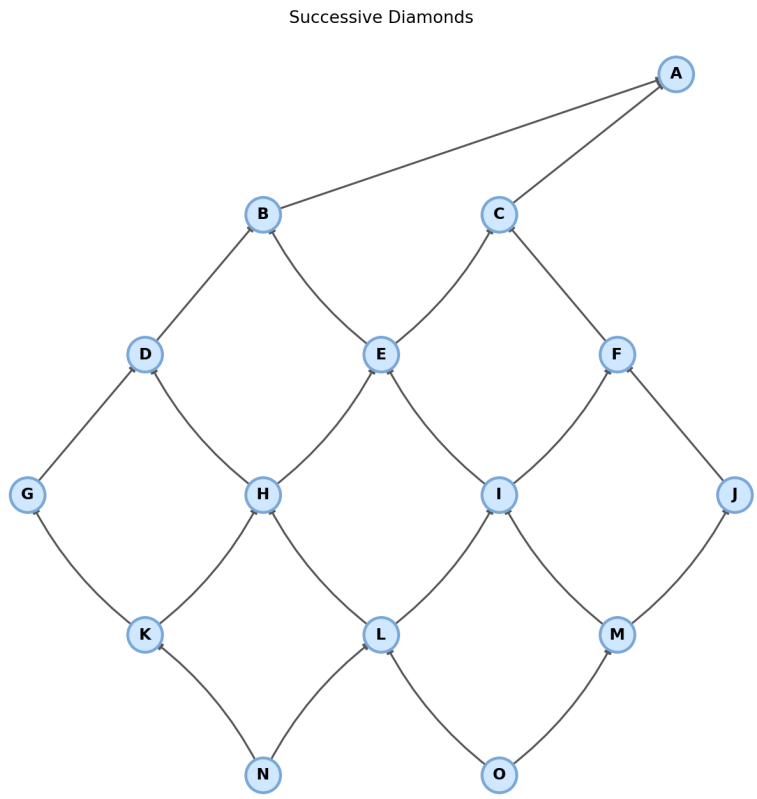

Figure 12: Successive Diamonds

Economic structure. A sequence of interconnected diamond-shaped modules where each stage of production splits into parallel subpaths and later reconverges before moving upstream. This layered symmetry generates multiple routes between distant nodes while preserving a clear global direction of flow.

Justification. The diamond architecture models production systems with repeated joint ventures or subcontracting arrangements, where parallel entities perform similar transformations before consolidation. Each merge–split cycle multiplies the number of potential pricing interfaces, creating a rich environment to test how firms can alternate between competition and collaboration to obscure value creation points and enable nuanced forms of profit-shifting across sequential stages.

Core metrics

Metric	Value
Number of Nodes	15
Number of Edges	22
Max Betweenness Centrality	0.075
Longest Path	5
Density	0.209

13 Tri-Cluster

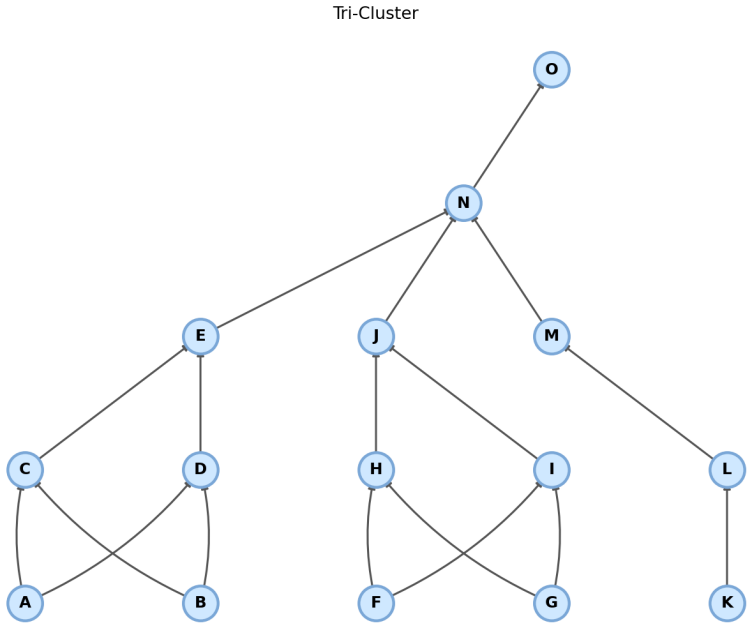

Figure 13: Tri-Cluster

Economic structure. Three independent production clusters—each internally dense and self-sufficient—operate in parallel before converging into a single coordination node that outputs the final good. Each cluster combines multiple low-level transformations before integration at the top.

Justification. Because each production branch contains complex internal transformations before convergence, firms may specialize in distinct subchains and later exchange semi-finished goods among them. This setup reproduces multi-division production groups where internal markets arise n4aturally, providing an ideal testbed for analyzing multi-layer profit-shifting and the emergence of hidden intra-group trade channels.

Core metrics

Metric	Value
Number of Nodes	15
Number of Edges	18
Max Betweenness Centrality	0.071
Longest Path	4
Density	0.171

14 Tri-Spine Convergent

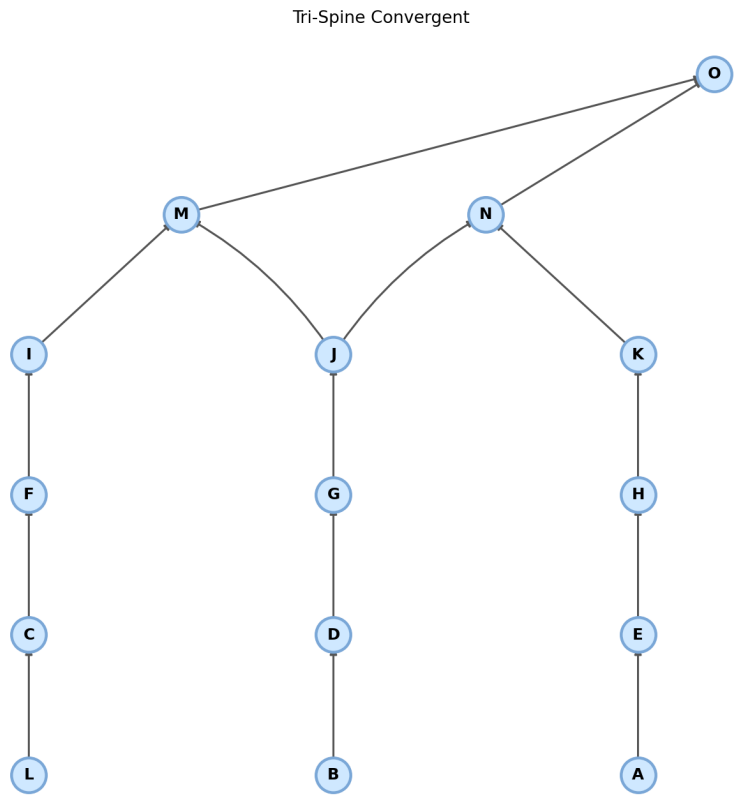

Figure 14: Tri-Spine Convergent

Economic structure. Three independent linear production chains operate in parallel before merging through two intermediate modules into a single final good. Each early-stage path can be managed autonomously, with interconnections appearing only at advanced stages of production.

Justification. Because the initial chains can be produced independently and later integrated through intermediate goods, firms may naturally exchange components across branches, enabling internal trade and price adjustments among related entities. This structure creates realistic opportunities for profit-shifting that are subtle and difficult to trace through conventional audit procedures.

Core metrics

Metric	Value
Number of Nodes	15
Number of Edges	15
Max Betweenness Centrality	0.049
Longest Path	5
Density	0.143

15 All-in-one

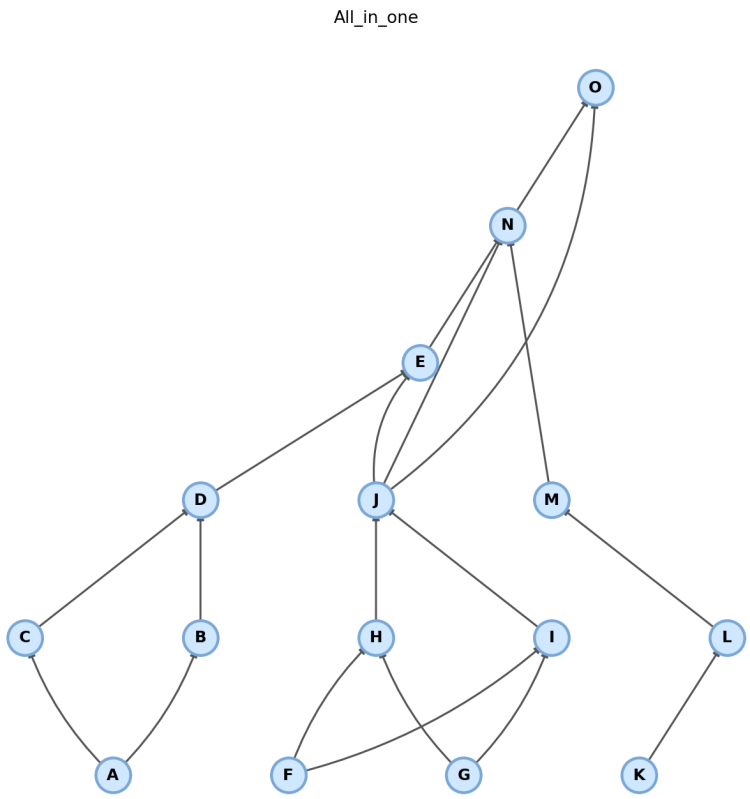

Figure 15: All-in-one

Economic structure. This graph integrates three semi-independent production branches that converge progressively into a common output. It combines linear sequences, diamond-shaped substructures, and cross-layer links, allowing intermediate goods to participate in multiple higher-level assemblies. This configuration blends hierarchical depth with parallel processing stages.

Justification. The coexistence of partially autonomous branches and shared intermediates creates flexible transaction pathways between firms operating in related stages. Such overlap allows cost reallocations and internal exchanges that can modify the apparent origin and value of goods, providing a nuanced environment for testing the algorithms' ability to identify profit-shifting patterns within complex production networks.

Core metrics

Metric	Value
Number of Nodes	15
Number of Edges	19
Max Betweenness Centrality	0.065
Longest Path	5
Density	0.18